

Strong and Weak Convergence of Euler–Maruyama and Milstein Schemes for Geometric Brownian Motion and a Non-Affine Stochastic Volatility Model

Project 1 (ODE IVP) — Direction 4: Financial SDE

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Abstract

[150–250 words answering: (1) what problem and why interesting — state that two models are compared, geometric Brownian motion as an exact benchmark and a non-affine stochastic volatility model with no closed-form transition law; (2) what methods you implemented — Euler–Maruyama and Milstein on the price process, Euler–Maruyama on the two-dimensional non-affine state, and a Brownian engine that produces one path shared by all step sizes; (3) key findings — the measured strong orders, the weak orders, whether the two weak curves coincide, and whether the increments preserve their moments under coarsening; (4) the main limitation — that the non-affine reference is numerical, so its rate is empirical, and that no scalar Milstein result applies there.]

Keywords: stochastic differential equations, Euler–Maruyama, Milstein method, strong convergence, weak convergence, mean-square stability, Itô–Taylor expansion, non-affine diffusion, Monte Carlo uncertainty, geometric Brownian motion

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1 Introduction and Motivation

1.1 Problem Background

1.1.1 SDEs in Finance

A deterministic ordinary differential equation produces a single future for a single initial condition. Asset prices do not behave in this way: two investors who hold the same stock on the same day will not see the same price path tomorrow. Randomness therefore has to enter the model itself, and the standard way to do this is to add a Brownian term to the differential equation. The result is a stochastic differential equation (SDE) of the general form

$$dX_t = a(X_t, t) dt + b(X_t, t) dW_t, \quad (1)$$

where a is the drift, b is the diffusion coefficient, and W_t is a standard Brownian motion. The two coefficients play different roles. The drift describes the average tendency of the process, while the diffusion describes how strongly random shocks are amplified. Their relative size determines how much of the observed variation is systematic and how much is noise.

The classical starting point is the geometric Brownian motion (GBM) model of Black, Merton and Scholes [?], in which both the drift and the diffusion are proportional to the current price:

$$dS_t = \mu S_t dt + \sigma S_t dW_t. \quad (2)$$

Because of its exponential structure, (2) can be solved in closed form, and its transition law is known exactly. This makes GBM an ideal *verification* problem: any code written for it can be compared against a reference that contains no discretisation error at all.

The same property, however, makes GBM unsuitable as a *modelling* problem. Constant volatility cannot reproduce two features that are easy to observe in real option markets: the implied volatility smile and the tendency of volatility to rise when prices fall. Modelling these effects requires the volatility itself to be a random process, which leads to the second model studied here. The report therefore uses two models with deliberately different status: GBM supplies an exact benchmark, while the stochastic volatility model is a problem in which numerical integration is genuinely part of the solution rather than a check of a known answer.

1.1.2 Itô's Lemma

Equation (2) is written in the price S_t , but it is more convenient to work with the log-price $X_t = \log S_t$. The chain rule of ordinary calculus cannot be used for this change of variable, because a Brownian path has infinite variation. The correct tool is Itô's lemma: for $f \in C^{1,2}$ and X_t satisfying (2),

$$df(X_t, t) = \left(\frac{\partial f}{\partial t} + a \frac{\partial f}{\partial x} + \frac{1}{2} b^2 \frac{\partial^2 f}{\partial x^2} \right) dt + b \frac{\partial f}{\partial x} dW_t. \quad (3)$$

The extra second-order term in (3) is the essential difference between stochastic and deterministic calculus.

Applying (3) with $f(s) = \log s$, $a = \mu S_t$ and $b = \sigma S_t$ gives

$$dX_t = \left(\mu - \frac{1}{2} \sigma^2 \right) dt + \sigma dW_t, \quad (4)$$

so the log-price satisfies an SDE with *constant* coefficients. The $-\frac{1}{2}\sigma^2$ term is not a modelling choice; it is forced by Itô's lemma, and dropping it changes the expected value of the price by a factor of $e^{\sigma^2 T/2}$. Integrating (4) gives the exact solution

$$S(T) = S_0 \exp \left[\left(\mu - \frac{1}{2} \sigma^2 \right) T + \sigma W(T) \right]. \quad (5)$$

Two consequences of (??) shape the whole numerical study, and they are made precise in Section ?? . Since the drift and the diffusion are constant, the transition law of X is exactly Gaussian, so the updating rule applied to X reproduces the exact solution at every grid point. At the same time, the constant diffusion means the noise is additive: the Milstein correction, which is proportional to the derivative of the diffusion coefficient, vanishes identically on X , and no strong order can be measured there. To obtain distinct orders, the price S itself must be discretised, where the noise σS does depend on the state.

We finally note the convention issue. The Itô form is used throughout, which is the standard convention in finance and is the reason the $-\frac{1}{2}\sigma^2$ correction appears above. The same convention produces the $-\frac{1}{2}g(Y_t)^2$ term in the stochastic volatility model introduced next.

1.1.3 Non-Affine Stochastic Volatility

The observation that motivates a stochastic volatility model is simple: implied volatility varies with strike and maturity, and it tends to increase when the underlying price falls. A constant σ cannot generate either pattern. The most widely used remedy is the Heston model [?], in which the variance V_t follows a mean-reverting square-root diffusion. Heston belongs to the class of *affine* diffusions: both its drift and its covariance are affine functions of the state, and consequently the characteristic function satisfies a Riccati ordinary differential equation that can be solved in closed form. Prices are then recovered by Fourier inversion. Affine structure is therefore not a technical detail; it is exactly what makes a model analytically tractable [?].

The model used in this report keeps the economic motivation but gives up that tractability. Writing $X_t = \log S_t$, its two equations are

$$dX_t = \left(\mu - \frac{1}{2}g(Y_t)^2\right) dt + g(Y_t) dW_t^{(1)}, \quad dY_t = \kappa(\theta - Y_t) dt + \xi\sqrt{1 + Y_t^2} dW_t^{(2)},$$

with a bounded logistic volatility function g and correlated Brownian motions. The factor Y_t is mean-reverting, so volatility is persistent rather than independent from period to period, and a negative correlation produces the leverage effect described above. The logistic function keeps the instantaneous volatility strictly between two positive bounds, which removes the vanishing- and negative-variance pathologies that arise in square-root models. Simulating X rather than S keeps the price positive by construction. The system is stated formally, with all parameter values, in Section ?? .

What makes this model *non-affine* is visible in the two coefficients. The diffusion of Y_t is $\xi\sqrt{1 + Y_t^2}$, which is not an affine function of Y_t , and the drift of X_t depends on the square of a logistic function. Both facts break the affine structure described above. The model therefore has no closed-form transition density, no closed-form characteristic function, and no practical exact sampler. This distinction should be stated precisely: the equations do have an explicit integral representation, but it is not a usable simulation formula. Time discretisation is consequently part of the model solution, and the numerical reference against which coarse schemes are measured must itself be validated by refinement, rather than taken from a formula.

1.2 Mathematical Models

This section fixes the notation and states the two models precisely. Section ?? defines the driving noise, Section ?? states the GBM benchmark, and Section ?? states the non-affine stochastic volatility model. All parameter values given here are the values used by the code described in Section 3, and both models are posed on the interval $0 \leq t \leq T = 1$.

1.2.1 Brownian Motion

Definition 1.1 (Standard Brownian motion). A real-valued process $\{W_t\}_{t \geq 0}$ on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$ is a *standard Brownian motion* if

- (i) $W_0 = 0$ almost surely;
- (ii) for every $0 \leq s < t$ the increment $W_t - W_s$ is independent of $\mathcal{F}_s$ and $W_t - W_s \sim \mathcal{N}(0, t - s)$;
- (iii) the map $t \mapsto W_t$ is continuous almost surely.

Three properties of this process are used repeatedly below. First, $\mathbb{E}[W_t] = 0$ and $\text{Var}[W_t] = t$, and $\{W_t\}$ is a martingale. Second, the quadratic variation of a Brownian path is $[W]_t = t$, which is why the ordinary chain rule fails and the second-order term in Itô's lemma appears. Third, Brownian paths are continuous but nowhere differentiable and have infinite total variation on every interval, so (??) cannot be read as a classical differential equation.

To couple two sources of randomness, we use a two-dimensional Brownian motion $\mathbf{W}_t = (W_t^{(1)}, W_t^{(2)})$ with

$$d\langle W^{(1)}, W^{(2)} \rangle_t = \rho dt, \quad |\rho| < 1. \quad (6)$$

Such a pair can always be built from two independent standard Brownian motions $B^{(1)}$ and $B^{(2)}$ by setting $W^{(1)} = B^{(1)}$ and $W^{(2)} = \rho B^{(1)} + \sqrt{1 - \rho^2} B^{(2)}$, which is the Cholesky factorisation of the correlation matrix. On a uniform grid $t_n = nh$ with $h = T/N$, this gives the increment convention used throughout the code:

$$\Delta W_n^{(1)} = \sqrt{h} Z_{1,n}, \quad \Delta W_n^{(2)} = \sqrt{h} (\rho Z_{1,n} + \sqrt{1 - \rho^2} Z_{2,n}), \quad (7)$$

with $Z_{1,n}, Z_{2,n} \sim \mathcal{N}(0, 1)$ independent. Each increment then has variance h , and $\text{Cov}(\Delta W_n^{(1)}, \Delta W_n^{(2)}) = \rho h$. The symbol h is used for the step size in this report; the corresponding variable in the code is named `dt`.

1.2.2 Geometric Brownian Motion

Notation and Parameters The single state variable is the price $S_t \in (0, \infty)$ itself, so the state space has dimension one. The problem pack does not prescribe GBM parameters, since GBM serves here only as a verification benchmark; the values below are therefore chosen explicitly and used in every GBM experiment.

Symbol	Meaning	Value	Unit
S_0	initial price	100	—
μ	drift rate	0.05	yr ⁻¹
σ	volatility	0.20	yr ^{-1/2}
T	final time	1	yr
h	step size, $h = T/N$	varies	yr

Table 1: GBM benchmark parameters and step size.

SDE System

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \quad S_0 = 100, \quad 0 \leq t \leq T = 1. \quad (8)$$

Both coefficients are proportional to S_t , so they are globally Lipschitz and grow at most linearly. The model is autonomous: neither coefficient depends explicitly on t .

Exact Solution, Transition and Moments The continuous-time solution was derived in Section ?? and is given by (?). Because the drift and the diffusion of the log-price (??) are constant, the transition law over one step is exactly Gaussian:

$$X_{n+1} | X_n \sim \mathcal{N}(X_n + (\mu - \frac{1}{2}\sigma^2)h, \sigma^2h), \quad X_n = \log S_n. \quad (9)$$

Sampling this transition gives the exact simulation scheme

$$X_{n+1} = X_n + (\mu - \frac{1}{2}\sigma^2)h + \sigma\sqrt{h}Z_n, \quad S_{n+1} = \exp(X_{n+1}), \quad (10)$$

with $Z_n \sim \mathcal{N}(0, 1)$ independent. Iterating (??) is equivalent to applying Euler–Maruyama to the log-price equation and exponentiating: the two are identical at every grid point, so (??) carries *no discretisation error*. This is what makes it the reference in the strong convergence study of Section ??, and it is also why no convergence order can be measured on the log-price scale.

The exact first and second moments follow directly from (??):

$$\mathbb{E}[S(T)] = S_0 e^{\mu T} = 105.1271, \quad \text{Var}[S(T)] = S_0^2 e^{2\mu T} (e^{\sigma^2 T} - 1) = 451.03. \quad (11)$$

The corresponding standard deviation is 21.24, which sets the Monte Carlo sample size needed to resolve a given bias in Section ?. The distribution of $S(T)$ is lognormal with parameters $(\log S_0 + (\mu - \frac{1}{2}\sigma^2)T, \sigma^2 T)$, and this is the law against which the simulated distribution is compared.

1.2.3 Non-Affine Stochastic Volatility

Notation and Parameters The state is the pair $z_t = (X_t, Y_t) \in \mathbb{R}^2$, where $X_t = \log S_t$ is the log-price and Y_t is a dimensionless volatility factor. The price is recovered as $S_t = e^{X_t}$, so $S_t > 0$ holds by construction. The parameters are those of the reproducible benchmark in the problem pack.

Symbol	Meaning	Value	Unit
S_0	initial price	100	—
Y_0	initial volatility factor	0	—
μ	drift rate of the log-price	0.05	yr ⁻¹
κ	mean-reversion speed of Y	2	yr ⁻¹
θ	long-run level of Y	-0.2	—
ξ	volatility-of-volatility	0.6	yr ^{-1/2}
ρ	correlation of $W^{(1)}, W^{(2)}$	-0.7	—
$\sigma_{\min}$	lower bound of g	0.10	yr ^{-1/2}
$\sigma_{\max}$	upper bound of g	0.50	yr ^{-1/2}
T	final time	1	yr

Table 2: Non-affine stochastic volatility benchmark parameters.

Two derived quantities are useful for orientation. Since g is logistic, $g(0) = (\sigma_{\min} + \sigma_{\max})/2 = 0.30$ is the instantaneous volatility at $t = 0$, while $g(\theta) = 0.2801$ is the level that volatility reverts to. Because $Y_0 = 0 > \theta$, the volatility factor starts above its long-run level and drifts downwards over the year.

SDE System

$$dX_t = (\mu - \frac{1}{2}g(Y_t)^2)dt + g(Y_t)dW_t^{(1)}, \quad (12)$$

$$dY_t = \kappa(\theta - Y_t)dt + \xi\sqrt{1 + Y_t^2}dW_t^{(2)}, \quad (13)$$

with the bounded logistic volatility function

$$g(y) = \sigma_{\min} + \frac{\sigma_{\max} - \sigma_{\min}}{1 + e^{-y}}, \quad 0 < \sigma_{\min} < \sigma_{\max}, \quad (14)$$

and the correlation (??). The initial condition is $(X_0, Y_0) = (\log 100, 0)$ and the interval is again $0 \leq t \leq T = 1$. The system is autonomous. The drift of X depends on Y only, so X_t does not feed back into Y_t ; the pair is one-way coupled.

Assumption 1.2 (Regularity of the coefficients). For both systems (??) and (??)–(??), the drift and diffusion coefficients are globally Lipschitz and satisfy a linear growth bound.

Assumption ?? is easy to verify for these two models, and the constants are small. For the logistic function, $g'(y) \leq (\sigma_{\max} - \sigma_{\min})/4 = 0.10$, so g is Lipschitz with constant 0.10 and bounded by 0.50. The drift of X depends on Y through $\frac{1}{2}g^2$, whose derivative is at most $gg' \leq 0.05$. The diffusion of Y satisfies $|\xi \sqrt{1 + y^2}| \leq \xi(1 + |y|)$, so it grows linearly with constant $\xi = 0.6$ and is Lipschitz with the same constant. Because these bounds hold uniformly, a unique strong solution of each system exists for all $t \geq 0$. Assumption ?? is exactly the hypothesis invoked by the convergence statements in Section ??, so it is stated here once and referenced there.

Remark 1.3 (An exact identity that survives). The boundedness of g gives one exact result even though the model is not tractable. Writing the price equation in the form $dS_t = \mu S_t dt + g(Y_t)S_t dW_t^{(1)}$, the discounted price $S_t e^{-\mu t}$ is a local martingale whose diffusion coefficient is bounded. Novikov’s condition therefore holds, since $\frac{1}{2} \int_0^T g(Y_u)^2 du \leq \frac{1}{2} \sigma_{\max}^2 T = 0.125$ is a deterministic bound, and consequently

$$\mathbb{E}[S_T] = S_0 e^{\mu T} = 105.1271. \quad (15)$$

This identity is exact and independent of $\kappa, \theta, \xi, \rho, \sigma_{\min}$ and $\sigma_{\max}$. It is used in Section ?? as a hard test of the implementation, because an error in the $-\frac{1}{2}g(Y_t)^2$ term of (??) would shift the sample mean away from 105.1271 by a detectable amount.

What the model does *not* provide is an analytically usable transition law. As explained in Section ??, the quadratic dependence of the variance on Y_t destroys the affine structure, so there is no closed-form transition density, no closed-form characteristic function and no practical exact sampler. Discretising (??)–(??) is therefore part of the solution of the model, and the reference used in Section ?? is a finely resolved numerical solution rather than a formula.

1.3 Overview of the Report

Section ?? derives the two discrete schemes, explains where their convergence orders come from, and discusses which of the usual ODE tools do and do not transfer to SDEs. Section ?? describes the implementation, the shared-path Brownian engine and the checks applied to the code before any result is quoted. Section ?? presents the numerical experiments, and Section ?? concludes.

2 Discretisation and Analysis

This section derives the two schemes used for GBM and the scheme used for the non-affine model, explains where their convergence orders come from, and states clearly which of the usual ODE tools do and do not transfer to SDEs. All statements below refer to the models of Section ?? and Section ??, and to Assumption ??.

2.1 Discrete Schemes

2.1.1 Euler–Maruyama

Consider the general scalar Itô SDE $dX_t = a(X_t)dt + b(X_t)dW_t$. Integrating over one step $[t_n, t_{n+1}]$ with $h = t_{n+1} - t_n$ gives

$$X_{n+1} = X_n + \int_{t_n}^{t_{n+1}} a(X_s) ds + \int_{t_n}^{t_{n+1}} b(X_s) dW_s. \quad (16)$$

The Euler–Maruyama (EM) scheme is obtained by freezing both integrands at the left endpoint:

$$X_{n+1} = X_n + a(X_n)h + b(X_n)\Delta W_n, \quad \Delta W_n = W_{t_{n+1}} - W_{t_n}. \quad (17)$$

The deterministic integral is approximated by a rectangle rule and the Itô integral by its integrand at the left endpoint; both approximations are consistent to first order.

Applied to the two models of this report, (??) gives

$$(\text{GBM, price}) \quad S_{n+1} = S_n + \mu S_n h + \sigma S_n \Delta W_n, \quad (18)$$

$$(\text{GBM, log-price}) \quad X_{n+1} = X_n + \left(\mu - \frac{1}{2}\sigma^2\right)h + \sigma\Delta W_n, \quad (19)$$

$$\begin{aligned} (\text{non-affine SV}) \quad X_{n+1} &= X_n + \left(\mu - \frac{1}{2}g(Y_n)^2\right)h + g(Y_n)\Delta W_n^{(1)}, \\ Y_{n+1} &= Y_n + \kappa(\theta - Y_n)h + \xi\sqrt{1 + Y_n^2}\Delta W_n^{(2)}. \end{aligned} \quad (20)$$

Equation (??) is the scheme implemented as `dX_rhs` in `SDEs/gbm.py`, and (??) as `dS_rhs`. Equations (??) are implemented as `dXdY_rhs` in `SDEs/nasv.py`.

2.1.2 Milstein

The Milstein scheme [?] keeps the first stochastic correction that EM discards. To derive it, expand the diffusion coefficient along the path,

$$b(X_s) = b(X_n) + b'(X_n)(X_s - X_n) + \mathcal{O}((X_s - X_n)^2), \quad (21)$$

and use the leading behaviour $X_s - X_n \approx b(X_n)(W_s - W_n)$, valid at the order required. The Itô integral in (??) then becomes

$$\int_{t_n}^{t_{n+1}} b(X_s) dW_s \approx b(X_n)\Delta W_n + b(X_n)b'(X_n) \int_{t_n}^{t_{n+1}} (W_s - W_n) dW_s. \quad (22)$$

The remaining Itô integral is elementary. Applying Itô's lemma (??) to $f(w) = \frac{1}{2}w^2$ gives $d(\frac{1}{2}W_s^2) = W_s dW_s + \frac{1}{2}ds$, hence

$$\int_{t_n}^{t_{n+1}} (W_s - W_n) dW_s = \frac{1}{2}(\Delta W_n^2 - h). \quad (23)$$

Substituting (??) into (??) and neglecting the higher-order terms gives the Milstein scheme

$$X_{n+1} = X_n + a(X_n)h + b(X_n)\Delta W_n + \frac{1}{2}b(X_n)b'(X_n)(\Delta W_n^2 - h). \quad (24)$$

The correction term is proportional to bb' , so it vanishes whenever the diffusion coefficient is constant.

For GBM on the price, $a(S) = \mu S$, $b(S) = \sigma S$ and $b'(S) = \sigma$, so $bb' = \sigma^2 S$ and

$$S_{n+1} = S_n + \mu S_n h + \sigma S_n \Delta W_n + \frac{1}{2} \sigma^2 S_n (\Delta W_n^2 - h), \quad (25)$$

which is `dS_correction` in `SDEs/gbm.py`. For GBM on the log-price, $b \equiv \sigma$ is constant, so the correction vanishes identically: Milstein and Euler–Maruyama coincide on X . This is the precise sense in which a strong order cannot be measured on the log-price scale, and it is why (??) rather than (??) is used in the order study of Section ??.

For the two-dimensional non-affine system, the same construction produces cross terms and iterated stochastic integrals,

$$z_{n+1} = z_n + a(z_n)h + B(z_n)\Delta W_n + \sum_{j_1, j_2} L^{j_1} L^{j_2} b(z_n) I_{(j_1, j_2), n}, \quad (26)$$

where B collects the diffusion vector fields b_1, b_2 and $I_{(j_1, j_2)}$ are iterated Itô integrals. The terms with $j_1 \neq j_2$ cannot be expressed through $\Delta W^{(1)} \Delta W^{(2)}$ unless the vector fields commute. For the non-affine model,

$$b_1(X, Y) = \begin{pmatrix} g(Y) \\ 0 \end{pmatrix}, \quad b_2(X, Y) = \begin{pmatrix} 0 \\ \xi \sqrt{1 + Y^2} \end{pmatrix}, \quad (27)$$

and a direct computation gives

$$[b_1, b_2] = \begin{pmatrix} -\xi \sqrt{1 + Y^2} g'(Y) \\ 0 \end{pmatrix} \neq 0, \quad (28)$$

because $g'(y) = (\sigma_{\max} - \sigma_{\min})e^{-y}(1 + e^{-y})^{-2} > 0$ everywhere. The noise is therefore non-commutative, the cross terms do not collapse, and a full Milstein method would require simulated Lévy areas. The scalar order-one result is not available for this system, so EM alone is used for the non-affine model throughout.

2.2 Consistency and Truncation Error

The Taylor expansion used above is incomplete, and the omitted terms set the orders. For the scalar case, the Itô–Taylor expansion to the first two stochastic orders is

$$X(t_{n+1}) = X_n + ah + b \Delta W_n + \frac{1}{2} bb' (\Delta W_n^2 - h) + R_n, \quad (29)$$

where the remainder R_n collects terms such as $\int \int b a' dW ds$ and the triple Itô integral. Table ?? lists the leading terms with their size in the $L^2(\Omega)$ norm and shows which scheme retains them.

Itô–Taylor term	Size in $L^2(\Omega)$	EM	Milstein
ah	$\mathcal{O}(h)$	✓	✓
$b \Delta W_n$	$\mathcal{O}(h^{1/2})$	✓	✓
$\frac{1}{2} bb' (\Delta W_n^2 - h)$	$\mathcal{O}(h)$	×	✓
remainder R_n	$\mathcal{O}(h^{3/2})$	×	×

Table 3: Leading Itô–Taylor terms for a scalar SDE, their L^2 size, and the schemes that retain them. Note that $\Delta W_n^2 - h$ has zero mean and variance $2h^2$, so the third term is of size $\mathcal{O}(h)$, strictly larger than the remainder.

Euler–Maruyama The local defect of EM is the omitted third term plus the remainder,

$$X(t_{n+1}) - X_{n+1}^{EM} = \underbrace{\frac{1}{2}bb'(\Delta W_n^2 - h)}_{\mathcal{O}(h)} + \underbrace{R_n}_{\mathcal{O}(h^{3/2})}, \quad (30)$$

so the local defect is of size $\mathcal{O}(h)$ in L^2 .

Milstein Milstein retains that term, so its local defect is just the remainder,

$$X(t_{n+1}) - X_{n+1}^{Mil} = R_n = \mathcal{O}(h^{3/2}). \quad (31)$$

How the local defects accumulate This step differs from the deterministic theory and is worth stating carefully. The leading local defect (??) has zero conditional expectation and is uncorrelated across steps, because it depends only on ΔW_n . Defects of this kind do not add up linearly; they add up like the increments of a random walk. Over $N = T/h$ steps the accumulated error is therefore of size

$$\sqrt{N} \times \mathcal{O}(h) = \mathcal{O}(h^{1/2}) \quad \text{for EM}, \quad \sqrt{N} \times \mathcal{O}(h^{3/2}) = \mathcal{O}(h) \quad \text{for Milstein.} \quad (32)$$

The factor $\sqrt{N}$ rather than N is the reason why a strong order of $1/2$ appears even though every local step is accurate to $\mathcal{O}(h)$. This is the main qualitative difference from the ODE case and it should not be smoothed over.

Drift-type defects behave differently: they are not mean-zero and do accumulate linearly, contributing $\mathcal{O}(h)$ after N steps. This distinction is what separates strong from weak convergence, and it is developed in Section ?? below.

2.3 Stability Analysis

The problem pack replaces the deterministic stability and adaptivity analysis of the ODE directions by convergence and uncertainty quantification for Direction 4 [?, ?]. This subsection is therefore not required by the brief. It is included for completeness and is correct, but its honest conclusion is that no stability constraint binds in either model.

2.3.1 Deterministic Reduction

Setting $\sigma = 0$ in (??) removes the noise. Both schemes (??) and (??) then reduce to the explicit Euler step $S_{n+1} = S_n(1 + \mu h)$, whose amplification factor is

$$R(z) = 1 + z, \quad z = \mu h, \quad (33)$$

with the classical absolute-stability region $|1 + z| < 1$. It should be noted that this region is not informative here: $\mu = 0.05 > 0$, so the exact solution grows and the notion of absolute stability, which describes decay, does not apply. The benchmark GBM problem contains no stiff mode.

2.3.2 Mean-Square Stability

For SDEs the natural analogue is mean-square stability. Consider the linear test equation $dX = \lambda X dt + \mu_d X dW$ with $\text{Re } \lambda < 0$, and define

$$R_{MS}(z, w) = \frac{\mathbb{E}[|X_{n+1}|^2]}{|X_n|^2}, \quad z = \lambda h, \quad w = \mu_d \sqrt{h}. \quad (34)$$

The scheme is mean-square stable if $R_{MS}(z, w) < 1$; the exact solution has $R_{MS}^{exact} = e^{2\lambda h + \mu_d^2 h}$.

Euler–Maruyama With $X_{n+1} = X_n(1 + \lambda h + \mu_d \Delta W_n)$ and $\xi = \Delta W_n / \sqrt{h} \sim \mathcal{N}(0, 1)$,

$$\mathbb{E}[|1 + z + w\xi|^2] = |1 + z|^2 + w^2,$$

because $\mathbb{E}[\xi] = 0$ and $\mathbb{E}[\xi^2] = 1$. Hence

$$R_{MS}^{EM}(z, w) = |1 + z|^2 + w^2. \quad (35)$$

Mean-square stability requires $|1 + z|^2 + w^2 < 1$, that is the disc $|1 + z| < \sqrt{1 - w^2}$, which is non-empty only if $w^2 < 1$, i.e. $\mu_d^2 h < 1$.

Milstein With the correction included, $X_{n+1} = X_n [1 + z + w\xi + \frac{1}{2}w^2(\xi^2 - 1)]$ and using the Gaussian moments $\mathbb{E}[\xi^2] = 1$, $\mathbb{E}[\xi^3] = 0$ and $\mathbb{E}[\xi^4] = 3$,

$$R_{MS}^{Mil}(z, w) = |1 + z|^2 + w^2 + \frac{1}{2}w^4. \quad (36)$$

The extra term $\frac{1}{2}w^4$ is easily lost by copying (??); its presence matters when w is not small. As a check, for $\lambda = 0$ the exact factor is $e^{w^2} = 1 + w^2 + \frac{1}{2}w^4 + \mathcal{O}(w^6)$, which agrees with (??) to the order shown.

Application to GBM GBM in the price is linear with $\lambda = \mu$ and $\mu_d = \sigma$, so $z = \mu h$ and $w = \sigma\sqrt{h}$. Since $\mu > 0$, the exact second moment grows like $e^{(2\mu + \sigma^2)h}$, and $R_{MS} < 1$ cannot be attained by any step size. Mean-square stability is therefore vacuous for this benchmark: there is no decay for a scheme to preserve. What remains meaningful is the *accuracy* of the second moment. For EM,

$$\mathbb{E}[S_N^2] = S_0^2 [(1 + \mu h)^2 + \sigma^2 h]^N, \quad (37)$$

whereas the exact value is $\mathbb{E}[S_T^2] = S_0^2 e^{(2\mu + \sigma^2)T}$. Comparing the two gives a relative deviation of size

$$(\mu^2 - \frac{1}{2}(2\mu + \sigma^2)^2)Th + \mathcal{O}(h^2) = -0.0073h + \mathcal{O}(h^2) \quad (38)$$

for the benchmark parameters. The deviation is of order h , which is exactly the weak order reported in Section ??, and for the Milstein scheme the same comparison improves by the term $\frac{1}{2}\sigma^4 h^2$.

The linearised non-affine system The Jacobian of the non-affine drift (??)–(??) is

$$J(z) = \begin{pmatrix} 0 & -g(Y)g'(Y) \\ 0 & -\kappa \end{pmatrix}, \quad (39)$$

because the drift of X depends on Y alone and X does not feed back. This matrix is triangular, so its eigenvalues are $\{0, -\kappa\} = \{0, -2\}$ at every state. The zero eigenvalue belongs to the neutral log-price direction, and the only negative eigenvalue imposes the step restriction $h < 2/\kappa = 1$, which is trivially satisfied by every step size used in this report. As in the GBM case, stability is not the binding constraint; accuracy is.

2.4 Convergence

Why the deterministic route is not available For an ODE one proves convergence by combining consistency with a stability estimate, and the Lax equivalence theorem closes the argument. No such theorem exists for SDEs. The reason is that the local defects of a stochastic scheme are random and their accumulation depends on their correlation structure, as Section ?? showed. The routes actually available are the Itô–Taylor expansion together with moment bounds, or the Gronwall argument in Kloeden and Platen [?]. The two orders are defined separately, and they are not interchangeable.

Definition 2.1 (Strong order). A scheme has strong order γ if, for both the scheme and the exact solution driven by the same Brownian path,

$$\mathbb{E}[|X_N - X(T)|] \leq Ch^\gamma \quad (40)$$

for all sufficiently small h . The error is a pathwise distance, so the two solutions must be coupled through common random numbers.

Definition 2.2 (Weak order). A scheme has weak order β if

$$|\mathbb{E}[f(X_N)] - \mathbb{E}[f(X(T))]| \leq Ch^\beta \quad (41)$$

for all f in a suitable class of smooth test functions.

Theorem 2.3 (Convergence of EM and Milstein, scalar case). *Under Assumption ??, Euler–Maruyama (??) has strong order 1/2 and weak order 1. If in addition $b \in C^1$ with Lipschitz derivative, Milstein (??) has strong order 1 and weak order 1.*

Theorem ?? is the reason for the two expectations used in Section ??: the strong error (??) for EM on the price is expected to fall like $h^{1/2}$, and for Milstein like h , while the weak error (??) is expected to fall like h for both. Note that Milstein improves the strong order but not the weak one. The added term has zero mean, so it does not change the bias; consequently the expected value of S_N is the same for EM and Milstein at every step size for GBM. This is a sharp and easily tested prediction, and it is used as a consistency check in Section ??.

Remark 2.4 (The benchmark is not a test of the theorem). On the log-price scale, EM reproduces the exact solution at grid points, so its error is identically zero rather than $\mathcal{O}(h^{1/2})$. Theorem ?? is still consistent, but it is not sharp for the degenerate reason discussed in Section ???. This is why the order study is carried out on the price process (??), where the noise is state-dependent and the theorem is informative.

Remark 2.5 (What can and cannot be measured for the non-affine model). For the non-affine system (??) no exact solution exists, so the strong error can only be computed against a finely resolved numerical solution. Two consequences follow. First, the order obtained this way is an empirical rate, not a theorem, and it must be reported as such. Second, because the noise (??) is non-commutative, EM is expected to show a strong order near 1/2 and no scalar Milstein comparison is available. The reference solution must itself be validated by refinement; agreement between two discretisations is evidence, not an exact error calculation.

3 Implementation

3.1 Algorithms

This section lists the algorithms that produce every number in Section ???. Two schemes are used for GBM and one for the non-affine model; the Brownian increments are produced once, outside the schemes, by the engine of Section ??, so that all step sizes and both methods share the same underlying path.

Algorithm 1 Euler–Maruyama (EM) for GBM, $dS = \mu S dt + \sigma S dW$

Input: initial price S_0 , step size h , steps N , increments $\{\Delta W_n\}_{n=0}^{N-1}$, drift μ , volatility σ

Output: approximate price S_N

```

1:  $S \leftarrow S_0$ 
2: for  $n = 0, 1, \dots, N - 1$  do
3:    $S \leftarrow S + \mu S h + \sigma S \Delta W_n$ 
4: end for
5: return  $S$ 

```

Algorithm 2 Milstein for GBM, $dS = \mu S dt + \sigma S dW$

Input: initial price S_0 , step size h , steps N , increments $\{\Delta W_n\}_{n=0}^{N-1}$, drift μ , volatility σ

Output: approximate price S_N

```

1:  $S \leftarrow S_0$ 
2: for  $n = 0, 1, \dots, N - 1$  do
3:    $S \leftarrow S + \mu S h + \sigma S \Delta W_n + \frac{1}{2} \sigma^2 S (\Delta W_n^2 - h)$ 
4: end for
5: return  $S$ 

```

Algorithm 3 Euler–Maruyama for the non-affine model, state $z_n = (X_n, Y_n)$

Input: input (X_0, Y_0) , step size h , steps N , correlated increments $\{\Delta W_n^{(1)}, \Delta W_n^{(2)}\}$, parameters $\mu, \kappa, \theta, \xi, \sigma_{\min}, \sigma_{\max}$

Output: X_N ; the price is $S_N = e^{X_N}$

```

1:  $X \leftarrow X_0, Y \leftarrow Y_0$ 
2: for  $n = 0, 1, \dots, N - 1$  do
3:    $\sigma_n \leftarrow g(Y)$  # logistic volatility, (??)
4:    $X \leftarrow X + (\mu - \frac{1}{2} \sigma_n^2) h + \sigma_n \Delta W_n^{(1)}$ 
5:    $Y \leftarrow Y + \kappa(\theta - Y) h + \xi \sqrt{1 + Y^2} \Delta W_n^{(2)}$ 
6: end for
7: return  $X$ 

```

No Milstein counterpart to Algorithm ?? is implemented. As shown in (??), the two diffusion vector fields of the non-affine system do not commute, so a multidimensional Milstein method would require cross terms and simulated Lévy areas. The solver therefore raises `NotImplementedError` if it is asked for a Milstein step on this model. This is deliberate: it makes it impossible to produce an order-one claim for the non-affine system by accident.

3.2 Brownian Engine and Shared Paths

Strong error is a pathwise distance (Definition ??), so the two solutions being compared must be driven by the same Brownian path. If instead a fresh path were drawn for each step size, the

differences between levels would be dominated by sampling noise rather than by the discretisation error, and no order could be estimated. The engine therefore works as follows.

Fine increments are generated once at the finest resolution $n_{\max}$, and the increments at any coarser resolution n with $n \mid n_{\max}$ are obtained by summing blocks of fine increments:

$$\left(\Delta W^{(c)}\right)_j = \sum_{i=jb+1}^{(j+1)b} \Delta W_i, \quad b = n_{\max}/n, \quad j = 0, \dots, n-1. \quad (42)$$

Because the coarse increments are block sums of the fine increments, the coarse path is exactly the fine path sampled on the coarse grid:

$$\sum_j \left(\Delta W^{(c)}\right)_j = \sum_i \Delta W_i = W(T) \quad \text{for every } n. \quad (43)$$

Equation (??) is what makes the strong error comparable across levels: the reference value $S(T)$ against which each scheme is measured is the same random variable at every resolution.

Two further properties of (??) matter. First, the correlation is preserved exactly: the coarse pair is still jointly Gaussian with $\text{Cov}(\Delta W^{(1)}, \Delta W^{(2)}) = \rho \Delta t$, so the parameter ρ is constant across the whole study even though the increments are never regenerated. Second, the nesting is transitive, so the levels can be produced by successive halving of the finest log. The implementation keeps only the current level in memory and discards the previous one, which bounds the memory by roughly twice the finest level rather than by the sum over all levels.

3.3 Code Structure

```
code/
|-- run_all.py                # runs every experiment, writes data/ and figures/
|-- SDEs/
|   |-- bm_engine.py         # standard_bm, correlated_bm_pair, extract_dw
|   |-- gbm.py               # dS_rhs, dS_correction, dX_rhs, S_exact
|   '-- nasv.py              # g, dXdY_rhs, benchmark parameters
|-- solvers/
|   |-- em.py                # em_step, em_solve (Euler-Maruyama)
|   '-- milstein.py          # milstein_step, milstein_solve (scalar GBM only)
|-- tools/
|   |-- mc.py                # Brownian simulator + increment-log I/O
|   '-- stats.py             # mean, standard error, CI, log-log slope fit
|-- experiments/
|   |-- exp0_simulate_bm.py   # shared-path smoke test
|   |-- exp1_strong_gbm.py    # strong convergence, EM vs Milstein
|   |-- exp2_weak_gbm.py     # weak convergence and the closed-form bias
|   |-- exp3_distribution.py  # distribution of S(T) vs the lognormal law
|   |-- exp4_positivity.py    # negativity rate of EM and Milstein
|   |-- exp5_strong_nasv.py   # coupled convergence against fine references
|   |-- exp6_payoff_nasv.py   # weak convergence of E[(S_T - 100)^+]
|   '-- exp7_diagnostics.py   # increment moments, finiteness, positivity
|-- data/                    # increment logs and result tables (CSV/JSON)
'-- figures/                  # figures used in this report
```

The two model modules contain only definitions; the two solver modules contain only stepping; the experiments contain only assembly and reporting. No experiment re-implements a scheme or draws its own noise. The solver state for the non-affine model is the pair (X, Y) rather than (S, Y) , so the price is always recovered as $S = e^X$ and positivity is structural rather than enforced by a projection.

3.4 Notation in the Code

Table ?? maps the symbols used in this report onto the names used in the code, so that every formula can be located in the repository without ambiguity.

Report symbol	Code name	Location
S_0	GBM_S0, NA_SV_S0	SDEs/gbm.py, SDEs/nasv.py
μ, σ	GBM_MU, GBM_SIGMA	SDEs/gbm.py
$\kappa, \theta, \xi, \rho$	NA_SV_KAPPA, NA_SV_THETA, NA_SV_XI, NA_SV_RHO	SDEs/nasv.py
$\sigma_{\min}, \sigma_{\max}$	NA_SV_SIGMA_MIN, NA_SV_SIGMA_MAX	SDEs/nasv.py
$g(\cdot)$	g	SDEs/nasv.py
h	dt	em_solve, milstein_solve
N	n_steps	em_solve, milstein_solve
$n_{\max}$	n_max	bm_engine, mc
ΔW_n	dW	bm_engine
$\Delta W_n^{(1)}, \Delta W_n^{(2)}$	dW1, dW2	bm_engine
$z_n = (X_n, Y_n)$	X, Y	solvers/em.py
S_N, X_N	out[:, -1]	em_solve, milstein_solve

Table 4: Report notation mapped to code identifiers.

3.5 Key Design Decisions

- **Shared Brownian path.** As explained in Section ??, the engine draws fine increments once at $n_{\max}$ and coarsens by summing blocks, so every step size and both methods see the same realisation. The increment logs are written to `data/` and reloaded by each experiment, so the coupling holds *between* experiments as well as within one. The cost is storage of order $2n_{\max}M$ doubles for the correlated pair, which is the reason Algorithm ?? is run in batches over paths rather than on the full ensemble at once.
- **Terminal values rather than trajectories.** The convergence and payoff estimators only need S_N , not the whole path, so the solvers are called in a mode that stores the current state and, where a trajectory is required for a figure, only for the small cases. This keeps the memory footprint at $\mathcal{O}(M)$ per level instead of $\mathcal{O}(Mn_{\max})$.
- **Vectorisation over paths, not over time.** The stepping loops run over the N time steps and operate on the whole path ensemble at once. For GBM this is not strictly necessary, because (??) and (??) can be written as cumulative products of per-step factors; this shortcut is used where the terminal value is all that is required. For the non-affine model it is necessary, because the coefficients of step $n + 1$ depend on the state reached at step n and no cumulative representation exists.
- **Log-price transform.** Discretising $X = \log S$ makes the GBM noise additive, so the exact sampler (??) needs no discretisation at all, and $S = e^X > 0$ holds automatically. For the non-affine model the same substitution keeps the price positive without a projection step, which would otherwise bias the weak error.
- **Milstein only for scalar GBM.** The correction (??) is implemented for the price process, where $b' \neq 0$. It is not implemented for the log-price (where it vanishes) or for the non-affine system (where the noise is non-commutative, (??)). The solver raises an error rather than returning a silently wrong order.
- **A separate exact reference.** The exact solution (??) is a distinct function, not a scheme, and it is never listed among the methods. This prevents the exact sampler from being reported as a numerical method with zero error.

- **Every estimate carries a standard error.** All Monte Carlo quantities are reported as $\hat{\theta} \pm 1.96 \hat{\text{se}}$, and a log–log slope is quoted only when the estimated difference between adjacent levels exceeds twice its paired standard error. Because the coarse and fine paths are coupled, that difference is computed on paired samples rather than on two independent batches.
- **Numerical evaluation of the logistic function.** The volatility (??) is evaluated as $1/(1 + e^{-y})$, which overflows for large negative y . The implementation uses a branch (or `expit`) that returns $\sigma_{\min}$ in that regime. Under the benchmark parameters Y remains of order one, so this is a robustness safeguard rather than an active concern.

3.6 Implementation Verification

Before any convergence result is quoted, the following checks are run. They are cheap, they are independent of each other, and each one catches a different class of error.

1. **Increment moments.** For the correlated pair, the sample variance of $\Delta W_n^{(1)}$ and $\Delta W_n^{(2)}$ should agree with h , and their sample correlation with ρ . This detects an error in the Cholesky step (??).
2. **Coupling identity.** For every $n \mid n_{\max}$, the cumulative sum of the coarse increments must equal the fine cumulative sum at the coarse grid points, to round-off. This detects an error in the block size or in the reshaping of the increment log.
3. **Exact moments for GBM.** The sample mean and variance of the exact sampler are compared with (??). For $M = 10^5$ paths the standard error of the mean is 0.067, so the sample mean must fall within about 0.13 of 105.1271.
4. **Exact mean for the non-affine model.** As shown in Remark ??, $\mathbb{E}[S_T] = 105.1271$ holds independently of every volatility parameter. This is the only exact identity available for that model and it is checked at every step size, because a sign or factor error in the $-\frac{1}{2}g(Y)^2$ term shows up immediately as a shift in the sample mean.
5. **Equality of the two weak curves.** By Theorem ?? and the zero mean of the Milstein correction, EM and Milstein must give the same $\mathbb{E}[S_N]$ for GBM at every step size. Any visible gap between the two weak-convergence curves indicates an implementation error, not a discretisation effect.
6. **Finiteness and positivity.** All simulated paths must be finite for both models, and $S = e^X > 0$ must hold throughout for the non-affine model.
7. **Reference resolution.** The numerical reference for the non-affine model is computed at two successive refinements, and the discrepancy between them must be smaller than the quantities being measured. Without this check the empirical rate of Remark ?? cannot be interpreted.

3.7 Reproducibility

Each experiment fixes its own generator seed, and the increment logs in `data/` are stored together with the seeds and parameters that produced them, so that every figure and table can be regenerated from `run_all.py` alone. Table ?? records the design of each experiment: the model, the methods compared, the step sizes, the number of paths and the cost metrics that are reported.

Script	Model	Methods	Step sizes	Paths
exp1	GBM	EM, Milstein, exact	$h = T/2^k, k = 4, \dots, 12$	10^5
exp2	GBM	EM, Milstein	same	10^5
exp3	GBM	exact, EM	$h = 1/252, 1/1024$	10^5
exp4	GBM	EM, Milstein	$h = 1/2^k, k = 2, \dots, 8$	10^6
exp5	non-affine	EM	$h = T/2^k, k = 4, \dots, 10$	10^4
exp6	non-affine	EM	$h = T/2^k, k = 3, \dots, 9$	10^4
exp7	both	—	—	10^4

Table 5: Design of the numerical experiments. The non-affine experiments use fine references at $h_{\text{ref}} = T/2^{13}$ and $T/2^{14}$; each experiment records the number of paths, the number of time steps, the wall time and the number of random numbers consumed.

4 Numerical Experiments

4.1 Verification

4.1.1 GBM with Exact Solution

[Present the exact-solution test. Report the sample mean and variance of the exact sampler against (??), with standard errors, and the moment checks of Section ?? . This establishes that the engine, the reference and the estimators agree before any order is quoted.]

4.1.2 Non-Affine Model: Numerical Reference and Nested-Grid Validation

[The reference here is a finely resolved numerical solution, not a manufactured solution, and it must be described as such. Report the reference at $h_{\text{ref}} = T/2^{13}$ and $T/2^{14}$, the discrepancy between the two, and the checks of Section ?? . State explicitly that agreement between two discretisations is evidence, not an exact error calculation.]

Step size h	Error	Std. error	Ratio	Order
$T/2^4$	[value]	[value]	—	—
$T/2^5$	[value]	[value]	[ratio]	[order]
$T/2^6$	[value]	[value]	[ratio]	[order]
$T/2^7$	[value]	[value]	[ratio]	[order]
$T/2^8$	[value]	[value]	[ratio]	[order]

Table 6: Convergence study for [test case], with M paths per level. A slope is quoted only for the levels where the difference between successive errors exceeds twice its paired standard error.

4.2 Strong Convergence

[Plot $\mathbb{E}|S_N - S(T)|$ against h on log-log axes for EM and Milstein on the price process, with error bars. Expected slopes: $1/2$ for EM and 1 for Milstein. Report the fitted slopes with confidence intervals and state the range of h over which each fit was made.]

4.3 Weak Convergence and Monte Carlo Uncertainty

[Report three separate pieces of evidence: the closed-form bias $\mathbb{E}[S_N] = S_0(1 + \mu h)^{T/h}$ compared with $S_0 e^{\mu T}$; a Monte Carlo estimate using common random numbers with confidence intervals; and the predicted equality of the EM and Milstein weak curves from Theorem ?? . For the payoff $\mathbb{E}[(S_T - 100)^+]$, use the numerical reference with the same random numbers and report a slope only when the estimated bias exceeds its sampling uncertainty.]

4.4 Distribution of $S(T)$ and Positivity

[Compare the simulated distribution of $S(T)$ with the lognormal law of Section ?? by histogram overlay and quantile plot. For the positive-preservation study, report the observed negativity rate of the EM and Milstein updates in the price against h , together with the analytic bound: the EM bracket $1 + \mu h + \sigma \Delta W_n$ is negative exactly when $\Delta W_n < -(1 + \mu h)/\sigma$, an event of probability $\Phi\left(-(1 + \mu h)/(\sigma\sqrt{h})\right) \sim e^{-1/(2\sigma^2 h)}$, while the Milstein bracket has minimum $\frac{1}{2} + (\mu - \frac{1}{2}\sigma^2)h$, so with the parameters of Table ?? it cannot become negative for any h . State which parameter values are needed to make the effect visible and that these are a stress test rather than a realistic setting.]

4.5 Cost versus Accuracy

[Report paths, time steps, wall time and random numbers consumed for each method and step size. Note that EM and Milstein consume the same random numbers, so the comparison is like for like.]

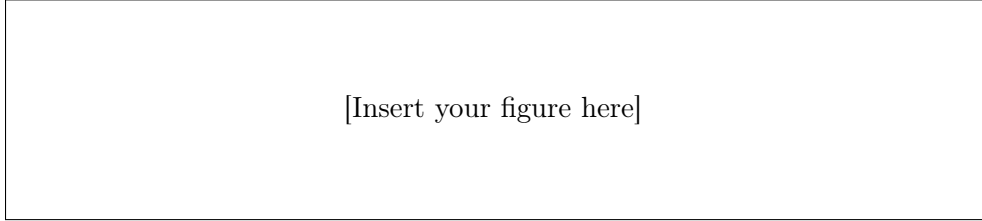


Figure 1: [Caption that interprets rather than describes. E.g. “The log–log plot gives slopes of 0.49 for Euler–Maruyama and 0.98 for Milstein, with the Milstein curve flattening above $h \approx 2^{-10}$ where the reference error becomes visible.”]

4.6 Limitations and Robustness

[Discuss the regimes where the conclusions weaken: very coarse steps, where the non-affine reference is insufficiently resolved; parameter values that make the correlated increments poorly conditioned; and the fact that no Milstein comparison exists for the non-affine model. This is not a weakness; it is scientific honesty.]

5 Conclusions

[Summarise in 3–4 bullet points:]

What you achieved.

ds in either model.

s rate is empirical.

oted option prices.

References

References

- [1] F. Black and M. Scholes, “The pricing of options and corporate liabilities,” *Journal of Political Economy* 81(3):637–654, 1973.
- [2] G. N. Milstein, “Approximate integration of stochastic differential equations,” *Theory of Probability and its Applications* 19(3):557–562, 1974.
- [3] P. E. Kloeden and E. Platen, *Numerical Solution of Stochastic Differential Equations*. Springer, 1992.
- [4] D. J. Higham, “An algorithmic introduction to numerical simulation of stochastic differential equations,” *SIAM Review* 43(3):525–546, 2001.
- [5] D. Duffie, D. Filipović and W. Schachermayer, “Affine processes and applications in finance,” *Annals of Applied Probability* 13(3):984–1053, 2003.
- [6] S. L. Heston, “A closed-form solution for options with stochastic volatility with applications to bond and currency options,” *Review of Financial Studies* 6(2):327–343, 1993.
- [7] C. R. Harris, K. J. Millman, S. J. van der Walt et al., “Array programming with NumPy,” *Nature* 585:357–362, 2020.

A AI Transparency Log

[Append the completed team AI Transparency Log here. See 06_Templates/ICS for the full form: tools used, significant claim audits, what you changed and why, and the intellectual ownership declarations.]

B Individual Contribution Statements

Each team member must complete the following (repeat for all 5 members):

Name: _____

1. My specific contributions to this project:

[3–5 sentences: which equations you derived, which code modules you wrote, which figures you produced, which report sections you drafted.]

2. The hardest conceptual obstacle I encountered:

[One specific mathematical or algorithmic difficulty and how you overcame it.]

3. One piece of AI-generated output my team used, and what I changed:

[Cross-reference an item from the AI Transparency Log.]

4. One thing I still do not fully understand:

[Not penalised; helps the instructor plan the next project.]

Signature: _____ **Date:** _____

C Supplementary Figures and Data

[Place any additional figures, tables, or code listings that would interrupt the main narrative but may be useful to a curious reader.]